

Module 4

Circuit Analysis in the S Domain

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Frequency response, Transfer function

So far, we were finding voltages and currents in a circuit with a constant frequency source.

If the amplitude of the sinusoidal source remain constant and vary the frequency, we obtain the circuit's frequency response.

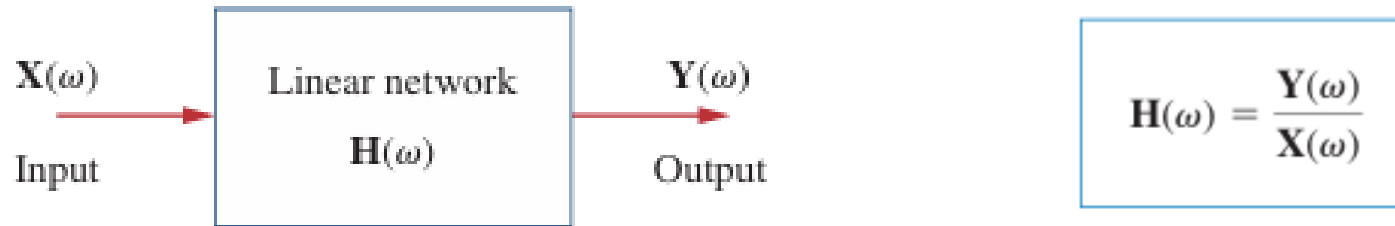
The frequency response may be regarded as a complete description of the sinusoidal steady-state behavior of a circuit as a function of frequency.

The frequency response of a circuit is the variation in its behavior with change in signal frequency.

The transfer function $H(\omega)$ (also called the network function) is a useful analytical tool for finding the frequency response of a circuit.

A transfer function is the frequency-dependent ratio of a forced function to a forcing function (or of an output to an input).

Frequency response, Transfer function



There are four possible transfer functions:

$$H(\omega) = \text{Voltage gain} = \frac{V_o(\omega)}{V_i(\omega)}$$

$$H(\omega) = \text{Current gain} = \frac{I_o(\omega)}{I_i(\omega)}$$

$$H(\omega) = \text{Transfer Impedance} = \frac{V_o(\omega)}{I_i(\omega)}$$

$$H(\omega) = \text{Transfer Admittance} = \frac{I_o(\omega)}{V_i(\omega)}$$

$H(\omega)$ has a magnitude $H(\omega)$ and a phase ϕ ; $H(\omega) = H(\omega) \angle \phi$.

Frequency response, Transfer function

The transfer function $\mathbf{H}(\omega)$ can be expressed in terms of its numerator polynomial and denominator polynomial as:

$$\mathbf{H}(\omega) = \frac{\mathbf{N}(\omega)}{\mathbf{D}(\omega)}$$

The roots of $\mathbf{N}(\omega) = 0$ are called the zeros of $\mathbf{H}(\omega)$ and are usually represented as $j\omega = \mathbf{z}_1, \mathbf{z}_2, \dots$. Similarly, the roots of $\mathbf{D}(\omega) = 0$ are usually represented as poles of $\mathbf{H}(\omega)$ and are represented as $j\omega = \mathbf{p}_1, \mathbf{p}_2, \dots$.

A zero, as a root of the numerator polynomial, is a value that results in a zero value of the function. A pole, as a root of the denominator polynomial, is a value for which the function is infinite

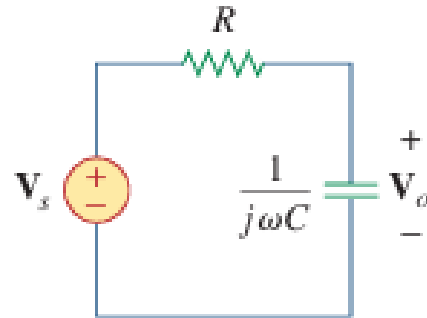
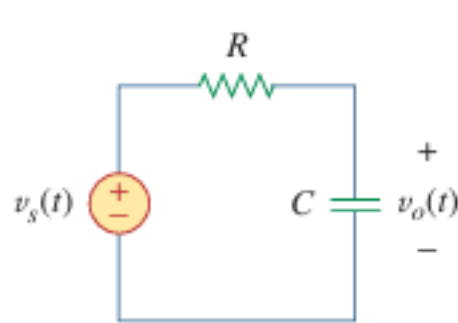
To avoid complex algebra, we can replace temporarily $j\omega$ with s when working with $\mathbf{H}(\omega)$ and replace s with $j\omega$ at the end.

Frequency response, Transfer function

1. For the RC circuit in figure, obtain the transfer function V_o/V_s and its frequency response.

Let, $v_s = V_m \cos \omega t$.

The frequency-domain equivalent of the circuit is



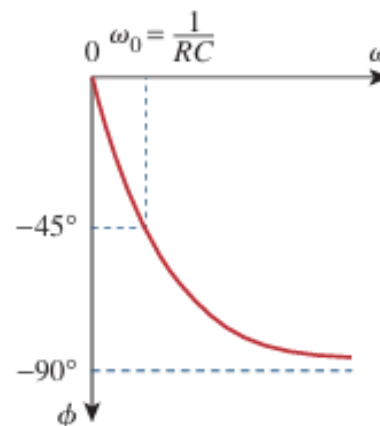
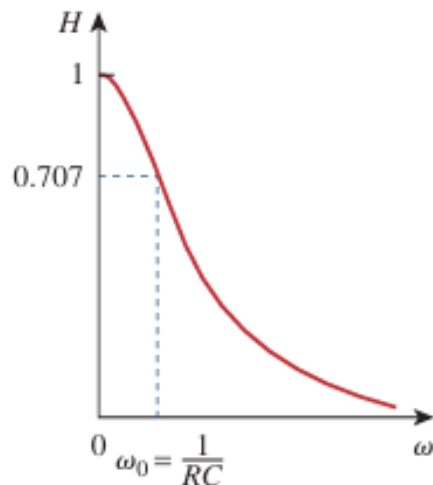
The magnitude and phase is

$$H = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}, \quad \phi = -\tan^{-1} \frac{\omega}{\omega_0}$$

where $\omega_0 = 1/RC$

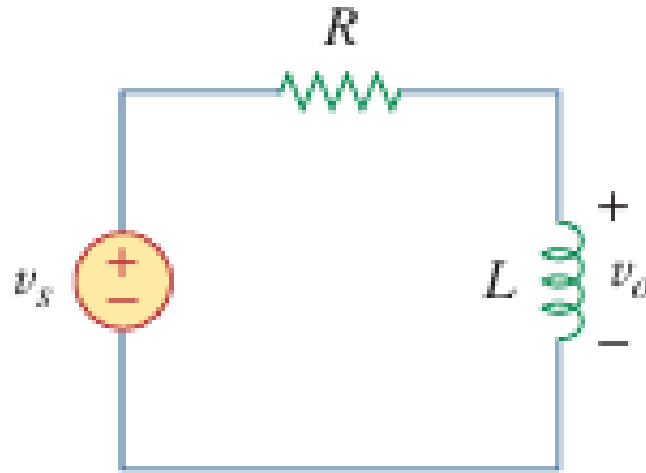
$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$$

At $\omega = 0$, $H = 1$ and $\phi = 0$. At $\omega = \omega_0$, $H = 1/\sqrt{2}$ and $\phi = -45^\circ$, At $\omega = \infty$, $H = 0$ and $\phi = -90^\circ$



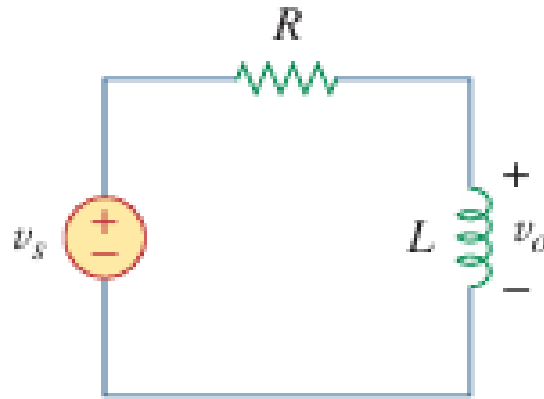
Frequency response, Transfer function

2. Obtain the transfer function of the RL circuit in figure, assuming $v_s = V_m \cos \omega t$. Sketch its frequency response.

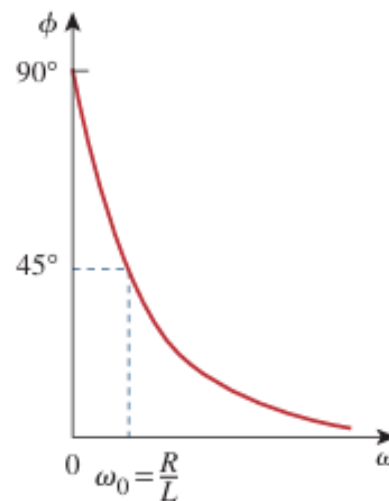
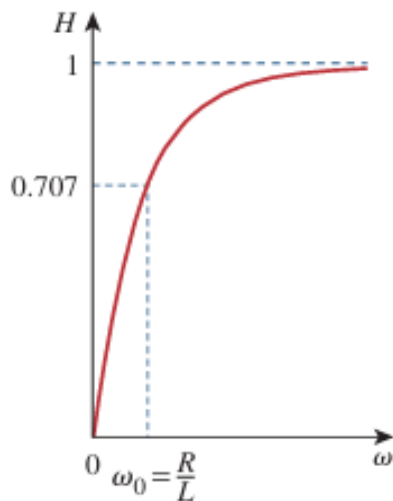


Frequency response, Transfer function

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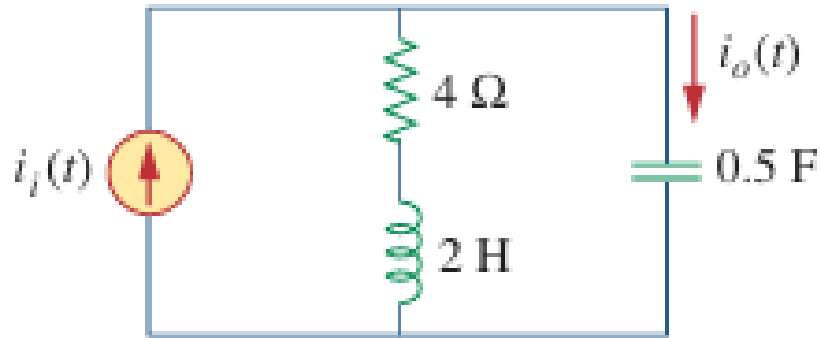


Answer: $j\omega L / (R + j\omega L)$



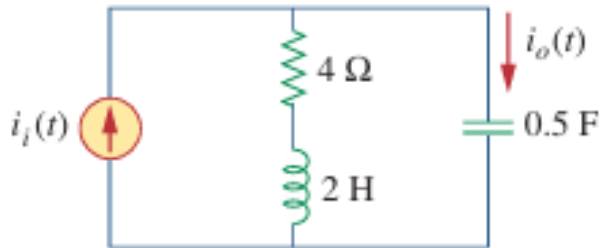
Frequency response, Transfer function

3. For the circuit in figure, calculate the gain $I_o(\omega)/I_i(\omega)$ and its poles and zeros



Frequency response, Transfer function

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By current division,

$$\mathbf{I}_o(\omega) = \frac{4 + j2\omega}{4 + j2\omega + 1/j0.5\omega} \mathbf{I}_i(\omega)$$

or

$$\frac{\mathbf{I}_o(\omega)}{\mathbf{I}_i(\omega)} = \frac{j0.5\omega(4 + j2\omega)}{1 + j2\omega + (j\omega)^2} = \frac{s(s + 2)}{s^2 + 2s + 1}, \quad s = j\omega$$

The zeros are at

$$s(s + 2) = 0 \quad \Rightarrow \quad z_1 = 0, z_2 = -2$$

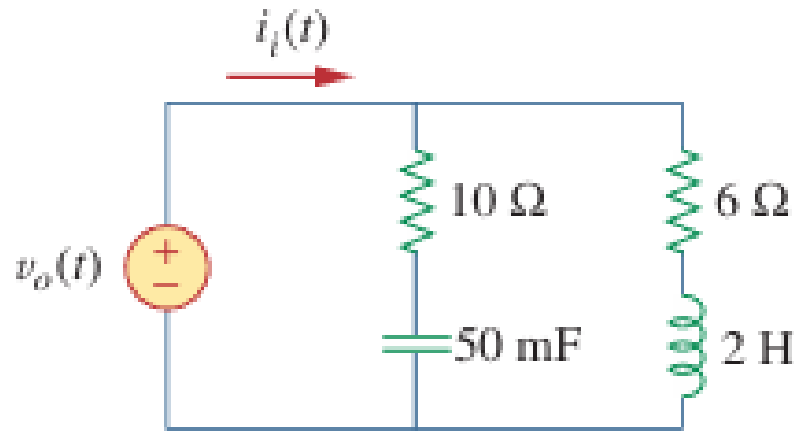
The poles are at

$$s^2 + 2s + 1 = (s + 1)^2 = 0$$

Thus, there is a repeated pole (or double pole) at $p = -1$.

Frequency response, Transfer function

4. Find the transfer function $V_o(\omega)/I_i(\omega)$ for the circuit in figure. Obtain its zeros and poles.



Answer: $\frac{10(s + 2)(s + 3)}{s^2 + 8s + 10}$, $s = j\omega$; zeros: $-2, -3$; poles: $-1.5505, -6.449$.

Next class: Laplace Transform